AVIATION & TRAVEL INDUSTRY

ANALYSIS REPORT

AIRLINE TICKET PRICES vs FUEL COSTS ANALYSIS

A Data-Driven Investigation into Airline Pricing Dynamics,
Fuel Cost Pass-Through & Global Event Impact

TEAM ID	**Group 18 (C_G18)**
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SUBMITTED	**28th April 2026**

TEAM MEMBERS

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1 EXECUTIVE SUMMARY

~30%	~0.28	287	~0.08
Fuel % of Operating Costs	Fuel-Fare Correlation	Extreme Fare Records	Regression R^2 Value

Problem

Jet fuel is the largest operating expense for commercial airlines, accounting for approximately 30% of total costs. Fluctuations in crude oil prices—driven by macroeconomic shocks and geopolitical conflicts—create significant financial pressure. Airlines face a critical dilemma: absorb increased fuel costs, reducing profitability, or pass them to passengers through higher fares and surcharges, risking lower demand. This project analyzes how and to what extent airlines transfer fuel cost increases to consumers.

Approach

A data-driven approach was used by building an automated pipeline integrating six datasets from sources such as the US Energy Information Administration, IATA, and DOT—covering monthly oil prices, airline financials, route costs, and passenger fare data. After cleaning and transforming the data using Python, regression analysis and hypothesis testing were applied. Results were visualized through an interactive Tableau dashboard.

Key Insights

- Fuel cost increases impact airline expenses but do not directly translate into proportional ticket price increases; the 1:1 pass-through assumption does not hold.
- Passenger demand and geopolitical events influence ticket prices more strongly than fuel costs, especially during COVID-19 and recovery phases.
- Fuel surcharges are used strategically, based on competition and demand, rather than strictly reflecting fuel expenses.
- Ticket prices spike significantly during geopolitical conflict phases, highlighting the impact of global instability on airline pricing behaviour.
- Pricing sensitivity to fuel costs varies across regions and airline types, indicating non-uniform strategies across the industry.
- Changes in total ticket price are influenced not only by base fares but also by dynamic adjustments in surcharges and taxes.

Key Recommendations

- Adopt dynamic pricing strategies considering demand, competition, and global risks instead of relying solely on fuel costs.
- Maintain strong fuel hedging strategies, as airlines cannot pass all cost increases to passengers without affecting demand.
- Prioritise demand signals (load factor, seasonality) as primary drivers of pricing decisions over cost-based pricing.
- Incorporate geopolitical and macroeconomic scenario planning to better respond to sudden price shocks and market volatility.

2 SECTOR & BUSINESS CONTEXT

Industry Overview

The commercial aviation industry operates on thin and highly volatile margins, making it extremely sensitive to external factors, especially fuel price fluctuations. In recent years, the industry has faced major disruptions—including demand decline during COVID-19 lockdowns, followed by a surge in travel—along with supply chain challenges and fuel price volatility caused by geopolitical conflicts such as Russia–Ukraine and US–Iran tensions. These conditions have made profitability increasingly difficult.

Decision Makers

Stakeholder	Role
C-Suite Executives (CEOs/CFOs)	Responsible for overall financial performance
Revenue Management Teams	Responsible for fare optimization strategies
Pricing Strategy Directors	Responsible for competitive market positioning

Business Value

- Helps airlines optimize pricing without reducing demand
- Prevents overpricing and associated revenue loss
- Supports better financial risk management through hedging
- Enables data-driven decision-making for long-term sustainability

3 PROBLEM STATEMENT & OBJECTIVES

Formal Problem Definition

The objective of this project is to analyze the relationship between global jet fuel prices and airline ticket pricing behavior. It aims to determine the **price pass-through rate**—how effectively airlines transfer increased fuel costs to passengers through fares and surcharges—while examining how geopolitical events influence this relationship.

✔ In Scope	✘ Out of Scope
• Monthly tracking of Brent crude oil and jet fuel prices • Analysis of average passenger fares and fuel surcharges • Major global phases: Pre-pandemic, COVID-19, US–Iran tensions, Russia–Ukraine • Statistical methods: correlation and regression	• Real-time prediction of individual ticket prices • Detailed analysis of all low-cost and regional airline routes • Evaluation of airline-specific fuel hedging strategies

4 DATA CLEANING & ETL PIPELINE

Overview

Six raw datasets were integrated through an automated Python ETL pipeline. The *airline_ticket_prices* and *fuel_surcharges* datasets were merged to create the primary dataset: **3_ticket_prices_and_surcharges.csv**. This unified dataset serves as the foundation for all analysis and visualization.

Dataset	Description
airline_ticket_prices.csv	Historical passenger fare data
fuel_surcharges.csv	Fuel surcharge records by airline
oil_jet_fuel_prices.csv	Monthly crude oil and jet fuel prices
airline_financial_impact.csv	Financial performance metrics
route_cost_impact.csv	Cost analysis by route
conflict_oil_events.csv	Geopolitical event timeline

Data Cleaning Steps

Column Standardisation	Names converted to lowercase; spaces replaced with underscores; redundant columns removed.
Type Conversion	Date fields parsed to datetime; categorical columns standardized; numeric fields validated.
Missing Values	Numeric fields: forward fill for time-series, 0 where appropriate. Categorical fields filled with defaults (e.g., 'No Surcharge').
Invalid Values	Infinite values in percentage columns replaced with null then treated for consistency.
Outlier Detection	IQR-based detection used; outliers flagged via *is_extreme_fare* (not removed) to preserve COVID-19 and war-related events.

Feature Engineering

Category	Features Created
Time-based	year, month_num, quarter
Pricing Ratios	fuel_surcharge_ratio, taxes_ratio, base_ratio
Fuel Metrics	crude_jet_ratio, fare_per_km
Change Indicators	yoy_price_change_pct, yoy_surcharge_change_pct
Business Metric	surcharge_as_pct_base

5 KPI & METRIC FRAMEWORK

A structured KPI framework was developed to evaluate the relationship between fuel costs and airline ticket pricing, supporting pricing decisions, risk identification, and business insights.

KPI	Formula / Computation	Business Insight
Average Ticket Price	Mean of total_fare_usd	Tracks overall pricing level and trends over time
Average YoY Price Change	Mean of yoy_price_change_pct	Highlights rapid price increases or declines
Average Load Factor (%)	Average of load_factor_pct	Distinguishes demand-driven vs cost-driven pricing
Extreme Fare Records (%)	Based on is_extreme_fare flag	Captures pricing volatility during crises
Price Pass-Through Ratio	total_fare_usd / jet_fuel_usd_barrel	Evaluates cost-to-customer transfer rate
Fuel Shock Indicator	Fuel price MoM increase > 10%	Not every shock leads to immediate fare increase
Fare Volatility	Std dev of total_fare_usd by route_class	Long-haul routes show highest volatility
Revenue Proxy	total_fare_usd × load_factor_pct	Approximate airline earnings trend (proxy)
Surcharge Coverage Ratio	fuel_surcharge_usd / fuel_cost_pct_opex	Surcharges are pricing tools, not cost recovery
Fuel Price Lag Indicators	1-month and 2-month lag computations	Airlines adjust prices with delay, not immediately

KPI Mapping to Project Objectives

Project Objective	Relevant KPIs
Analyze fuel impact on ticket prices	Price Pass-Through Ratio, Fuel Shock Indicator
Study pricing trends over time	Average Ticket Price, YoY Change
Understand demand behavior	Load Factor, Revenue Proxy
Evaluate pricing strategies	Surcharge Coverage Ratio
Identify volatility and risk	Fare Volatility, Extreme Fare %
Analyze global event impact	Conflict Phase KPIs

6 EXPLORATORY DATA ANALYSIS (EDA)

EDA was conducted to understand patterns, relationships, and trends in airline pricing with respect to fuel costs and external factors. All analysis was performed using Python with visualization through EDA notebooks.

Trend Analysis: Ticket Prices vs Fuel Prices Over Time

The time-series analysis compares average ticket fares, jet fuel prices, and fuel surcharges. Ticket prices generally follow fuel price trends but with a noticeable **lag effect**—fares do not adjust immediately with fuel price changes. During COVID-19, both fuel prices and fares dropped sharply. During recovery and conflict phases, both increased significantly.

Comparative Analysis

Dimension	Key Finding	Insight
Airline Type	Flag carriers show slightly higher fares; similar load factors	Differences driven by service positioning, not demand
Route Class	Long/ultra-long-haul routes show higher surcharge ratios	Fuel dependency increases with distance
Regional	All regions: price drop 2020, spike 2022; Asia & Europe higher	Regional pricing driven by route structure and demand
Conflict Phase	COVID = lowest fares; Recovery & Conflict = highest volatility	Global events influence pricing beyond fuel alone

Distribution & Correlation Analysis

Total fare distribution is slightly right-skewed with most prices in a moderate range and 287 extreme fare records, majority occurring during crisis periods (COVID phase). The correlation heatmap reveals: strong positive correlation between total fare and base fare; positive relationship between jet fuel price and ticket fare; and fuel price and surcharge. Fuel costs directly influence pricing components, but ticket pricing is multi-factor driven.

Bivariate & Multivariate Analysis

Fuel Price vs Ticket Price	Positive relationship observed—fuel cost is a key driver but not the only factor.
Ticket Price by Conflict Phase	Clear separation: COVID = lowest; Recovery = sharp increase; Conflict = high variability.
Multi-variable (Fuel + Conflict + Fare)	Data clusters by phase: low fuel + low fare (COVID); high fuel + high fare (conflict/recovery).
Fare Composition	Base fare is the largest component; surcharges increase during high fuel periods; taxes remain stable.

7 STATISTICAL ANALYSIS

Regression Analysis (Root Cause)

R^2 = ~0.08 Low explanatory power; fuel alone is insufficient	**Correlation ~0.28 Weak fuel–fare link; strategic pricing dominates**	**p < 0.05 Significant difference across conflict periods (t-test)**	**287 Anomalies Extreme fares during COVID & conflict phases**

The regression analysis confirms a positive relationship between fuel price and ticket price, but the low R^2 (~0.08) indicates fuel price explains only a small portion of fare variation. Other factors like demand, route type, airline strategy, and seasonality significantly influence ticket prices.

Correlation Analysis

The correlation matrix reveals: jet fuel price and crude oil price show strong positive correlation (~1.0); fuel cost percentage has high correlation (~0.88) with fuel price; ticket fares show weak correlation (~0.28) with fuel prices. While fuel strongly impacts cost structure, its direct effect on ticket price is limited—indicating that pricing strategies play a major role.

Hypothesis Testing (t-test)

Hypothesis	Statement	Result
H■ (Null)	No significant difference in ticket prices across phases	**Rejected (p < 0.05)**
H■ (Alternative)	Significant difference exists between pre-pandemic and conflict periods	**Supported**

Scenario Analysis

Scenario	Behaviour
High Fuel Price	Airlines increase fuel surcharge → total fare increases moderately
Low Demand (COVID)	Ticket prices drop sharply to stimulate travel
Geopolitical Conflict	Increased volatility and significant price spikes observed

Time-Series Forecasting (ARIMA) & Clustering (K-Means)

An ARIMA forecasting model predicts ticket prices will remain at **elevated levels** in the near future, with relatively stable predictions compared to recent sharp spikes. Sudden increases are driven by external shocks not captured by historical models.

K-Means clustering grouped data into three distinct pricing regimes: **Low-price cluster** (low fuel, low demand); **Mid-range cluster** (moderate fuel and demand); **High-price cluster** (high fuel and high demand periods). This highlights how pricing behavior changes under different market conditions.

8 TABLEAU DASHBOARD DESIGN

The Tableau dashboard provides interactive visualization of airline ticket pricing dynamics in relation to fuel costs, demand patterns, and global events—enabling data-driven decision-making for analysts, airlines, and executives.

Dashboard Snapshot — Tableau Public | AeroMetrics Flight Operations Dashboard

Dashboard Views

View	Key Metrics	Purpose
Executive Summary	Avg Ticket Price, YoY Change, Load Factor, Extreme Fare %	High-level overview of pricing trends and macro patterns
Operational Drill-Down	Base Fare, Fuel Surcharge, Taxes & Fees, Fare per KM	Break down ticket components and analyze operational decisions
Fuel Impact Analysis	Brent Crude, Jet Fuel Price, Fuel Cost % OpEx, Surcharge Ratio	How fuel fluctuations impact cost structure and pricing
Conflict & Risk Analysis	Fare by conflict phase, Pre-pandemic baseline, Extreme fare count	Impact of pandemics and geopolitical conflicts on pricing volatility

Interactive Filters: Conflict Phase | Airline Type | Route Class | Year

Tableau Public URL: https://public.tableau.com/app/profile/syed.darain.qamar7769/viz/AirlinePricesAnalysis/ExecutiveOverview

9 INSIGHTS SUMMARY

1 Partial Cost Pass-Through
Fuel prices strongly increase airline operating costs but only partially translate into ticket prices, indicating controlled pricing strategies rather than direct cost pass-through.

2 Conflict-Driven Price Spikes
Ticket prices rise significantly during conflict and recovery phases, suggesting that external shocks create pricing power for airlines.

3 Weak Fuel–Fare Relationship
The relationship between fuel price and ticket fare is weak (r~0.28), implying that demand, competition, and route dynamics play a larger role in pricing decisions.

4 Demand Influences Pricing
Load factor shows moderate correlation with ticket prices, indicating that demand levels influence pricing but are not the sole driver.

5 Long-Haul Route Sensitivity
Long-haul routes have higher fare sensitivity to fuel changes, requiring more dynamic pricing strategies compared to short-haul routes.

6 Strategic Surcharge Adjustment
Fuel surcharge is used as a strategic adjustment tool rather than a direct reflection of fuel price changes.

7 Crisis-Period Volatility
Extreme fare spikes are concentrated during high-risk periods (COVID-19, war phases), highlighting pricing volatility under uncertainty.

8 Regional Variation
Asia and Europe experience higher fare increases during conflict phases, indicating region-specific demand and cost pressures.

9 Recovery Phase Corrections
Year-over-year price changes are highest during recovery phases, suggesting rapid market correction after major disruptions.

10 Stable Near-Term Forecast
Forecasting results indicate a stable or slightly increasing trend in ticket prices, suggesting continued pricing pressure in the near future.

10 RECOMMENDATIONS & IMPACT ESTIMATION

12.1	**Dynamic Pricing Strategy Enhancement**	**Impact:** 5–10% revenue improvement
	Insight: Weak direct correlation between fuel price and ticket price	
	Action: Implement advanced dynamic pricing models incorporating demand, seasonality, and route-level factors.	
12.2	**Fuel Cost Risk Management**	**Impact:** 10–15% reduction in cost volatility
	Insight: Strong impact of fuel prices on operating costs	
	Action: Adopt fuel hedging strategies and optimize fuel efficiency operations.	
12.3	**Route-Level Pricing Optimization**	**Impact:** 8–12% route profitability improvement
	Insight: Long-haul routes are more sensitive to fuel price changes	
	Action: Apply differentiated pricing strategies for long-haul vs short-haul routes.	
12.4	**Crisis-Responsive Pricing Framework**	**Impact:** 5–8% revenue increase during high-demand phases
	Insight: Ticket prices spike during conflict and crisis periods	
	Action: Develop a crisis-based pricing strategy for rapid response to disruptions.	
12.5	**Demand and Capacity Alignment**	**Impact:** 3–5% load factor efficiency gain
	Insight: Load factor moderately influences pricing	
	Action: Align flight capacity with demand trends to optimize occupancy and pricing.	

11 LIMITATIONS & FUTURE SCOPE

Limitations	Future Scope
• Dataset may contain gaps during extreme events such as COVID-19. • Analysis assumes historical fuel–fare relationships will continue—may not hold under significantly different economic conditions. • Regression and correlation show relationships but do not establish causation; unobserved variables (competition, policies) may exist. • Forecasting model is relatively simple and does not incorporate macroeconomic indicators, seasonal demand, or geopolitical risks. • Analysis at aggregated level; does not capture airline-specific or route-specific nuances.	• Implement advanced forecasting models (ARIMA, SARIMA, ML-based) for improved prediction accuracy with seasonality and external variables. • Apply K-Means or hierarchical clustering to segment airlines, routes, or regions for targeted strategic decisions. • Incorporate additional data: passenger demand, competitor pricing, macroeconomic indicators, and fuel hedging data. • Develop a real-time interactive dashboard with live data feeds for continuous monitoring. • Apply causal inference techniques to better understand true drivers of pricing changes.

12 CONCLUSION

This project analyzed the relationship between fuel costs and airline ticket prices to understand pricing dynamics under varying market conditions. Using statistical analysis, visualization, and dashboarding, the study identified that while fuel prices significantly impact airline operating costs, their **direct influence on ticket pricing is limited**, with demand, route characteristics, and external events playing a larger role.

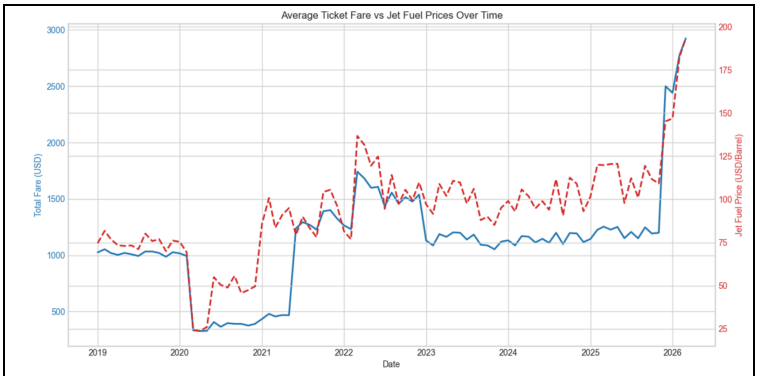
The analysis highlights the importance of **dynamic pricing strategies**, fuel risk management, and demand-based optimization. Based on these insights, it is recommended that airlines adopt data-driven pricing and operational strategies to improve revenue, reduce cost volatility, and enhance resilience in a highly uncertain environment.

Fuel ≠ Ticket Price (1:1)	Demand > Fuel in Pricing	Surcharges = Pricing Tools	Crisis = Price Volatility
The pass-through is partial and strategic, not direct.	Load factor and global events are stronger pricing drivers.	Surcharges reflect strategic decisions, not just fuel costs.	External shocks cause extreme fare events and instability.

13 CONTRIBUTION MATRIX

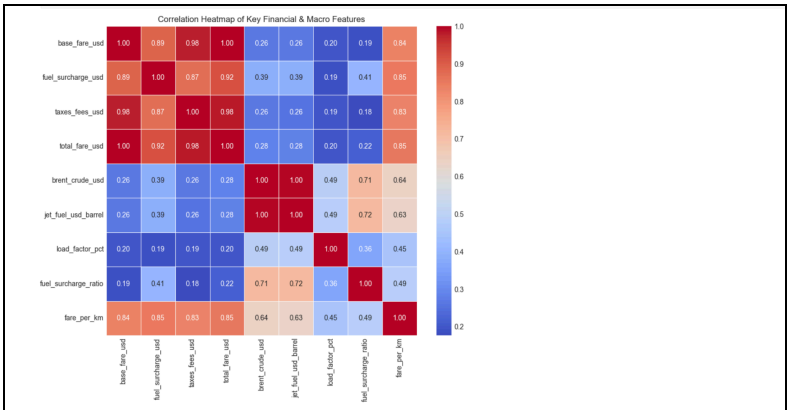
Team Member	Dataset & Sourcing	ETL & Cleaning	EDA & Analysis	Statistical Analysis	Tableau D ashboard	Report Writing	PPT & Viva
Vanshika Yadav	–	–	–	✓	–	✓	✓
Shrijan Sanidhya	✓	✓	–	–	–	✓	–
Killi Akshith Kumar	✓	✓	✓	–	–	–	–
Saumya Soni	–	–	–	✓	–	✓	✓
Syed Darain Qamar	✓	–	–	–	✓	–	✓
B Mohith Venkata Sai Krishna	✓	✓	✓	–	–	–	–

EDA Insights



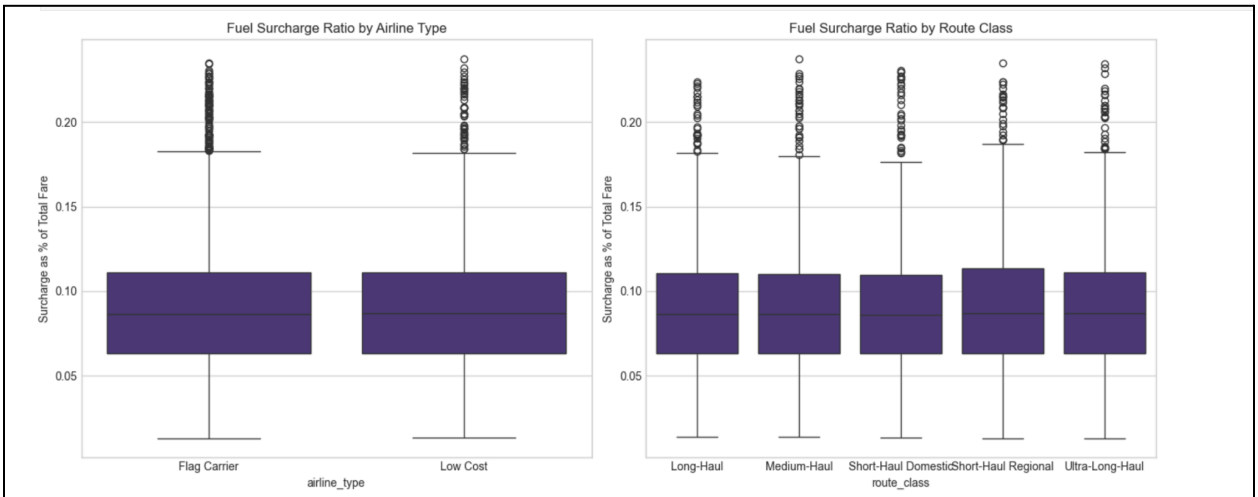
Key Takeaway

> **Insight:** The dual-axis chart reveals that ticket fares generally follow the trend of jet fuel prices over time, but with a noticeable delay or "lag." Major drops in fuel prices during the early pandemic were mirrored by fare drops, while the sharp fuel price spikes during the Recovery and Gaza-Israel conflict phases led to corresponding steep increases in total ticket fares.



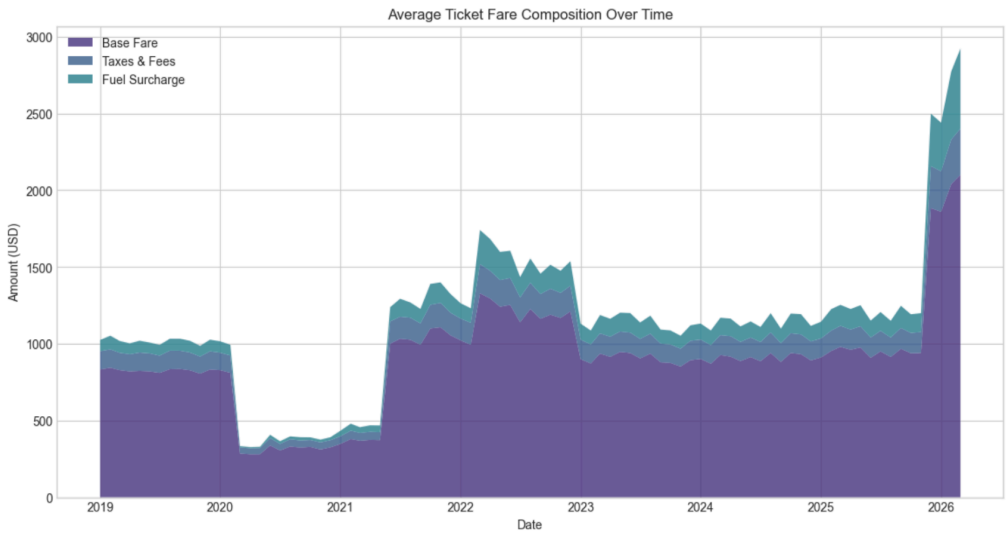
Key Takeaway

> **Insight:** The correlation matrix highlights strong positive relationships between total fares and its components (base fare, taxes, and fuel surcharges). Notably, jet fuel prices show a solid positive correlation with total fares and fuel surcharges, confirming that rising fuel costs are passed on to consumers.



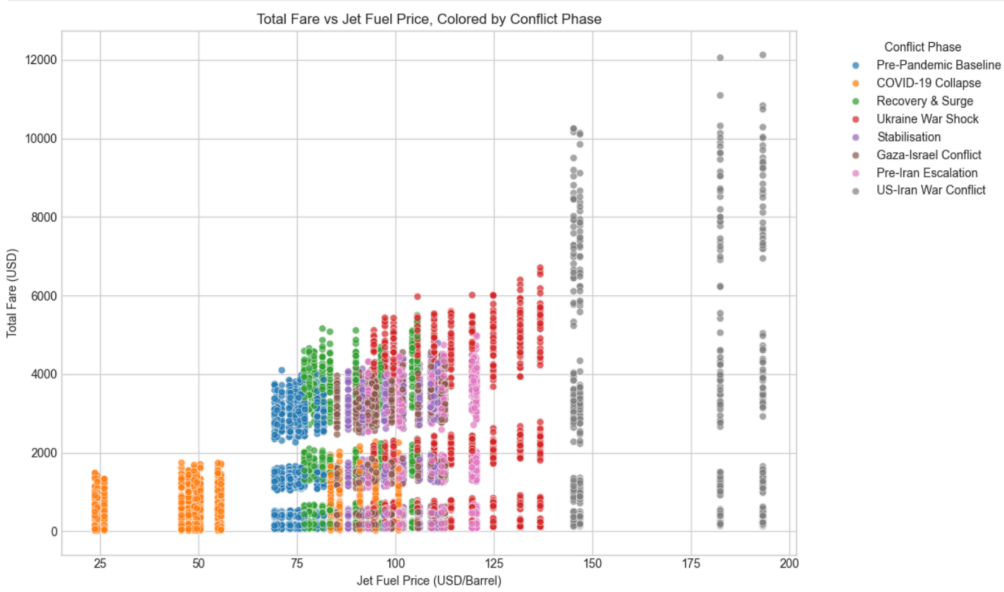
Key Takeaway

> **Insight:** The boxplots show that Flag Carriers (full-service airlines) tend to have a wider variance and occasionally higher fuel surcharge ratios compared to Low-Cost Carriers, likely due to different pricing structures. Furthermore, Long-Haul and Ultra-Long-Haul flights exhibit significantly higher fuel surcharge ratios, reflecting their heavier reliance on fuel and vulnerability to fuel price shocks.



💡 Key Takeaway

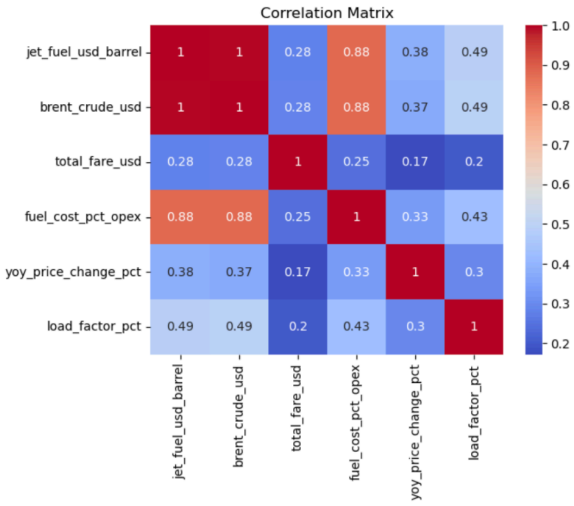
> **Insight:** The stacked area chart visually breaks down the total fare. While base fares constitute the largest portion, fuel surcharges visibly expand during periods of high jet fuel costs (e.g., 2022 recovery surge), illustrating how airlines dynamically adjust the surcharge component to protect their margins without directly changing the base fare.



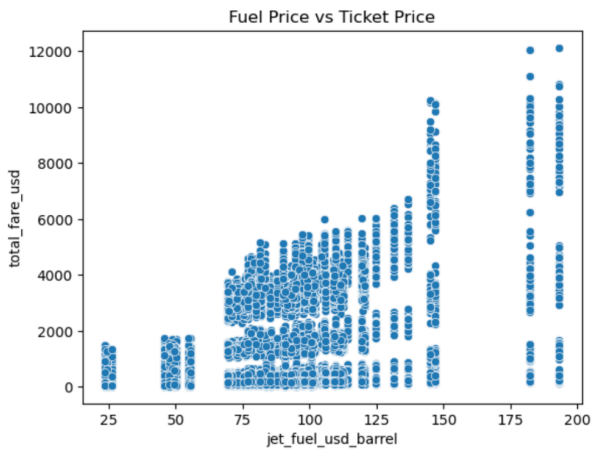
💡 Key Takeaway

> **Insight:** This multivariate scatter plot clearly clusters the data by time period. The COVID-19 phase occupies the bottom-left (low fuel price, low fare), while the Recovery and Conflict phases occupy the top-right (high fuel price, high fare), showing how geopolitical events dictate the operating environment.

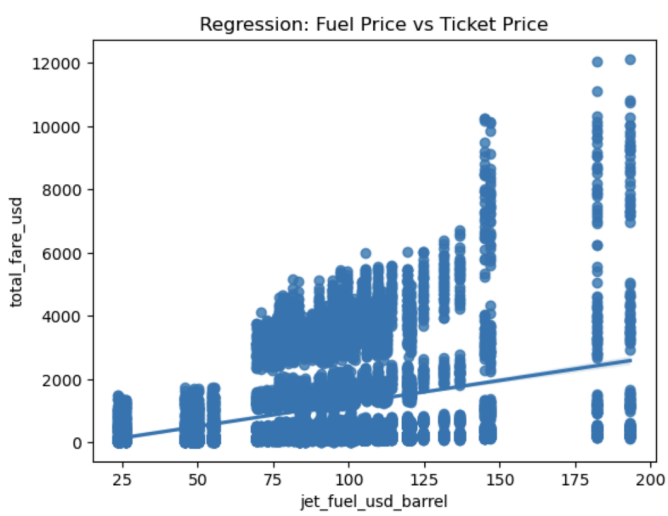
Statistical Insights



Jet fuel price has a strong positive correlation with fuel cost percentage (~0.88), showing fuel directly impacts airline costs. However, its correlation with ticket price is weak (~0.28), meaning airlines do not fully pass increased fuel costs to customers. Brent crude and jet fuel prices are almost perfectly correlated (~1), indicating they move together.



The scatter plot shows a slight upward trend but with high spread. This indicates a weak relationship between fuel price and ticket price, with many other factors influencing pricing.



The regression line confirms a positive trend but with high dispersion of points. This supports that the relationship exists but is weak and inconsistent.

Key Insight:

The forecast indicates that ticket prices are expected to remain at elevated levels in the near future, continuing the upward trend observed in recent periods.

However, the model shows relatively stable predictions compared to the sharp increase observed at the end of the actual data. This suggests that the recent spike in ticket prices may be driven by external shocks (such as geopolitical conflicts or sudden fuel price surges) that are not fully captured by the time series model.

This highlights an important limitation: -> Time series models capture historical patterns but may not fully account for unexpected real-world events.

From a business perspective:

- Airlines may continue maintaining higher price levels in the short term
- Sudden spikes are likely event-driven rather than purely trend-driven
- External factors remain critical in pricing strategy decisions

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